

Week 12 — 🎨 Turtle Art

Name: _____ Date: _____

*Spirograph recipe: **draw a shape, turn a little, repeat** — it closes perfectly when the turns total **360°**. And `colors[i % 6]` makes six colors paint forever.*

Today's words

Word	What it means
<code>pencolor()</code> / <code>bgcolor()</code>	the pen's ink / the paper's color
<code>speed(0)</code>	NO speed limit — the turtle teleports
spiral	<code>forward(i)</code> — every side one step longer than the last
spirograph	a shape rotated again and again until the turns fill 360°
<code>colors[i % 6]</code>	% wraps the count (0 1 2 3 4 5 0 1...) so the list cycles

1 • Match the code to the art 🎨

Write the letter of each snippet next to its picture (each is used once).

- | | |
|--|---|
| A) <code>for i in range(4):</code>
<code>forward(100)</code>
<code>right(90)</code> | B) <code>for i in range(100):</code>
<code>forward(i)</code>
<code>right(90)</code> |
| C) <code>for i in range(5):</code>
<code>forward(100)</code>
<code>right(144)</code> | D) <code>for i in range(36):</code>
<code>(draw a square)</code>
<code>right(10)</code> |

- ____ a five-pointed star
- ____ one plain square
- ____ squares spiraling outward, each side longer than the last
- ____ a round flower/mandala made of 36 tilted squares

2 • The rotation table 🔄

Repeats to close = **360 ÷ tilt**. Fill in the blanks.

tilt after each shape	repeats to close
10°	36
20°	_____
30°	_____
45°	_____
120°	_____

3 • Design your own spirograph 🖍️ (we run these NEXT WEEK!)

- Shape sides (3-8): ____ • Tilt angle: ____° • Repeats to close: ____
- Colors (pick up to 6): _____
- Sketch your prediction:



🧠 Brain teaser (optional — take it home)

Neat spirographs need tilt angles that **divide 360 evenly**. 1, 2, 3, 4, 5, and 6 all work... 7 doesn't! How many divisors of 360 can you find? (There are **24**.) Bonus pondering: ancient astronomers *chose* 360 for the circle — why was that such a convenient number to pick?